STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Is displacement scalar or vector?
2. x-component at angle theta?
3. Horizontal acceleration of an ideal projectile?
4. Direction of centripetal acceleration?
5. Magnitude of perpendicular components 3 and 4?
6. Relation between v and omega?
7. Ideal maximum-range angle at equal heights?
8. Vertical velocity at projectile peak?

Section B 2-mark questions - 6 x 2 = 12 marks

9. Resolve 10 N at 60 deg .
10. Displacement after a full circular lap?
11. v=12 m/s, r=6 m: a_c?
12. Horizontal launch from 20 m, g=10: flight time?
13. Perpendicular vectors 6 and 8: resultant?
14. rpm in rad/s?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Projectile at 20 m/s, 30 deg , g=10: T,H?
16. Forces 8 N east and 6 N north: resultant?
17. Cyclist at 10 m/s, radius 25 m: a_c?
18. m/s at 37 deg , sin=.6, cos=.8, g=10: time/range?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Projectile 20 m/s at 30 deg : time, height, range?
20. kg at 4 m/s in radius 2 m: inward force and work?